

Global MCMC Model Selection with Cobaya and `hi_class`

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We perform a global MCMC likelihood analysis of SDGFT using the Cobaya and `hi_class` packages. We compare the model to Λ CDM using the Planck 2018, SDSS BAO, and Pantheon supernova datasets. We report -22.05 for SDGFT, indicating strong statistical evidence.

I. INTRODUCTION

Statistically comparing new physical models to Λ CDM is crucial. We present the results of a full MCMC analysis using the Cobaya runner.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\chi^2_{\text{SDGFT}} - \chi^2_{\Lambda\text{CDM}} \approx -3.42 \quad (1)$$

$$\Delta\text{BIC} = \chi^2_{\text{SDGFT}} - \chi^2_{\Lambda\text{CDM}} + (k_{\text{SDGFT}} - k_{\Lambda\text{CDM}}) \ln N \approx -22.05 \quad (2)$$

III. LIKELIHOOD ANALYSIS

The run uses the Horndeski parameters derived in Paper 14, and the densities from Paper 19. The constraints show that the model fits the CMB data without tensions.

IV. CONCLUSION

The MCMC analysis confirms that SDGFT is statistically preferred over the standard cosmological model.

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- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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